המרכז
לחינוך סייבר
CYBER EDUCATION CENTER

Assembly for Reverse Engineering

Stack, Procedures

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The Stack

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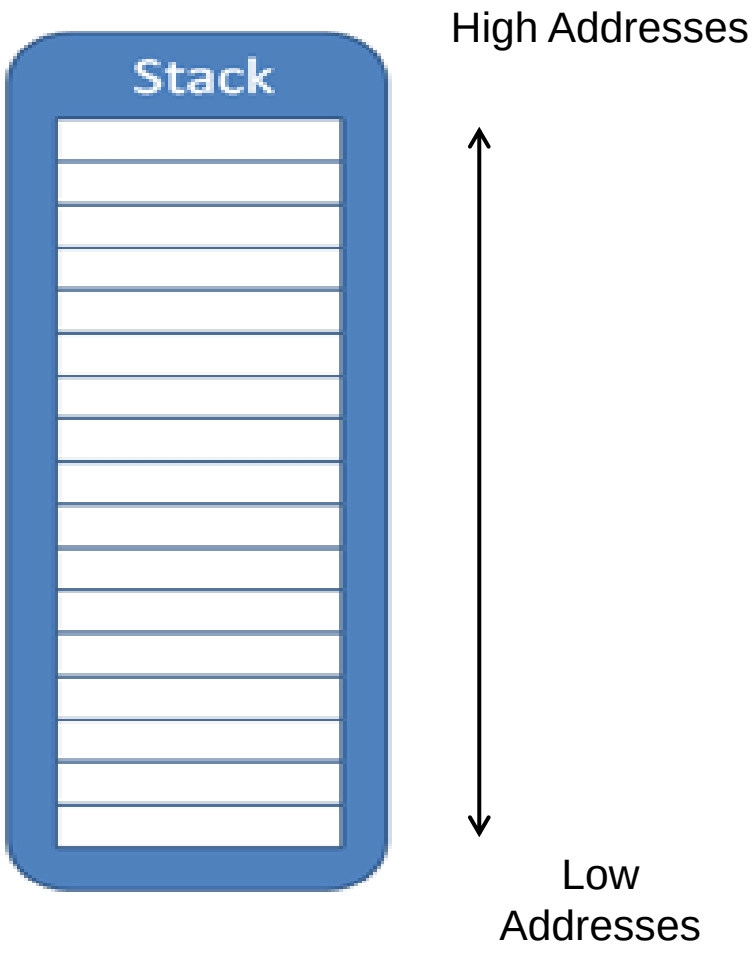
- Used for short term storage of:
  - Data
  - Addresses
- Important for understanding functions
- FASM default – 64K



Stack Register- ESP

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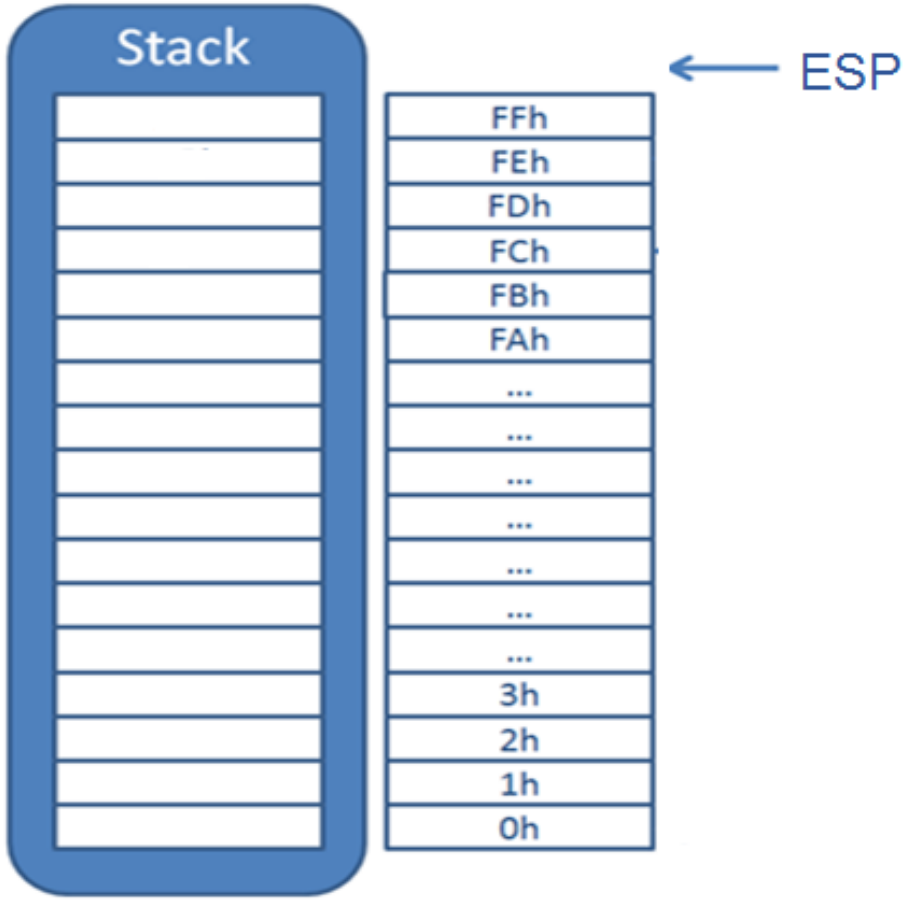
- ESP – (Extended) Stack Pointer.
- Initial value – stack size (yes- outside the stack)
- Example: stack size 100h



Inserting/ Extracting Stack Values

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- LIFO – Last In First Out
- Before inserting value to the stack – ESP decreases
- After extracting value from the stack – ESP increases
- PUSH inserts value
- POP extracts value

PUSH

- ESP is reduced by operand size
  - 4 or 2
  - 1 – not possible
- Value is stored in [ESP]

push operand

PUSH - Example

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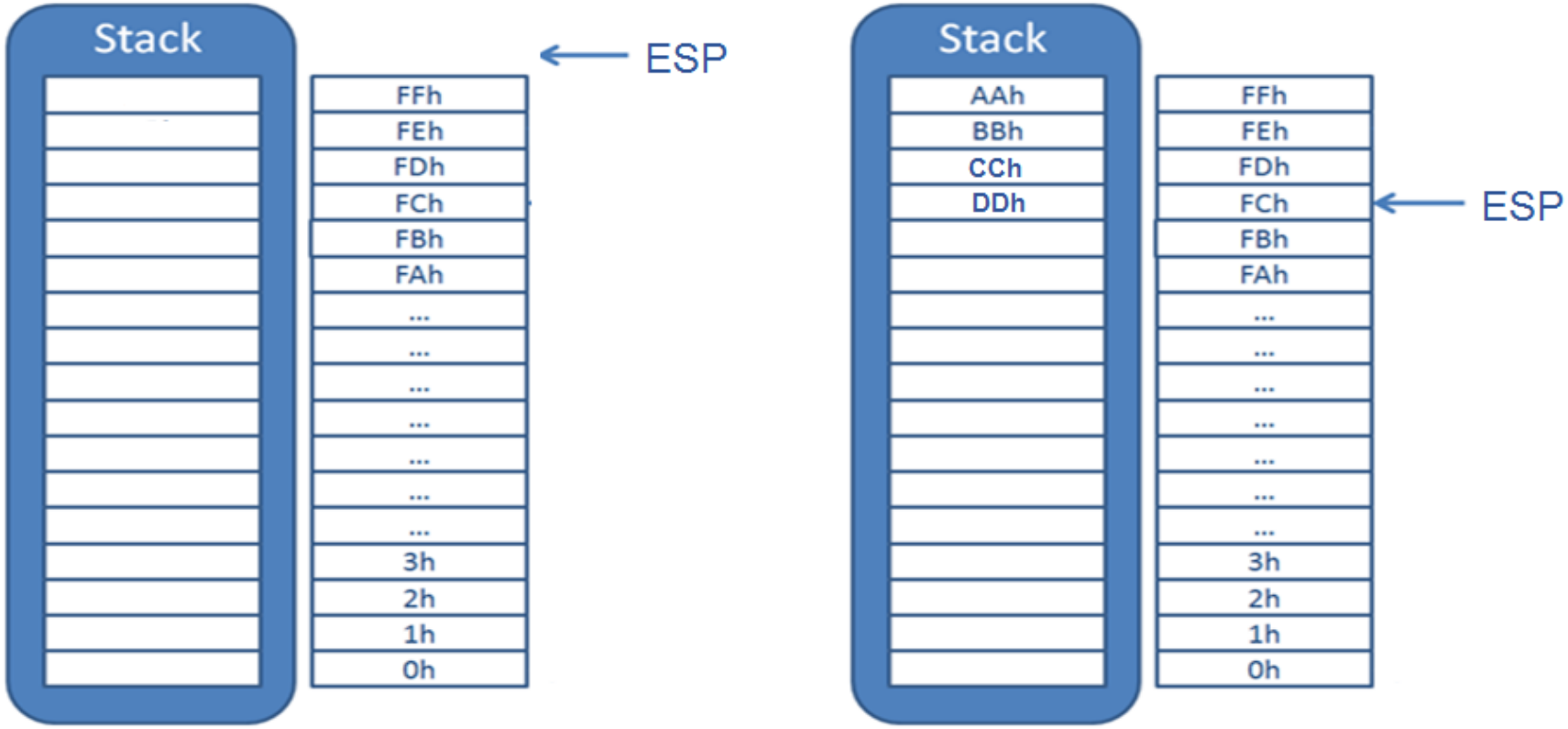
```
mov     eax, 0xAABBCCDD
push    eax
```



POP

- Value from [ESP] is copied to operand
- ESP is increased by operand size
  - 4 or 2

pop operand

Practice

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- Copy EAX to EBX without using MOV
- Add instructions so EAX == 4, EBX == 5

```
mov     eax, 4
mov     ebx, 5
push    eax
push    ebx
xor     eax, eax
xor     ebx, ebx
; add code to restore eax, ebx
```

Procedures

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- CALL, RET- what do they do?
- Self study
  - Note- debug with “Step Into”
  - Monitor EIP, ESP
- Challenge – using one PUSH, create infinite loop ☺

```
start:
        mov     eax, 4
        mov     ebx, 5
        mov     ecx, 6
        mov     edx, 7
        call    zero_regs
        push    0
        call    [ExitProcess]

zero_regs:
        xor     eax, eax
        xor     ebx, ebx
        xor     ecx, ecx
        xor     edx, edx
        ret
```

CALL- RET

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- CALL:
  - Decrease ESP
  - Copy the return address to the stack
  - Change EIP to start of procedure
  - Jump to EIP
- RET:
  - Read the return address, copy to EIP
  - Increase ESP
  - Jump to EIP

Class Exercise

- Mul.asm
- Fib.asm

Passing Parameters

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- Suppose we have a variable:

```
section '.data' data readable writeable
    I    dd   5
```

- Memory:

Address	Hex dump
00401000	05 00 00 00

- We can pass the variable to a procedure:
  - By value
  - By reference

Passing Parameters

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Pass by value

- Push value

```
push        [I]
```

- Stack:

```
000CFF88|| 00000005
```

- Procedure may not change parameter

Pass by reference

- Push address

```
push        I
```

- Stack:

```
000CFF88|| 00401000
```

- Procedure may change parameter

Passing Parameters

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- The procedure ‘simple’ accepts I,J,K as params, and returns EAX = I + J - K

```
section '.data' data readable writeable
    I    dd   5
    J    dd   6
    K    dd   7
section '.text' code readable executable

start:
    push    [I]
    push    [J]
    push    [K]
    call    simple
    push    0
    call    [ExitProcess]
```

- Can you write ‘simple’?

Accessing Parameters

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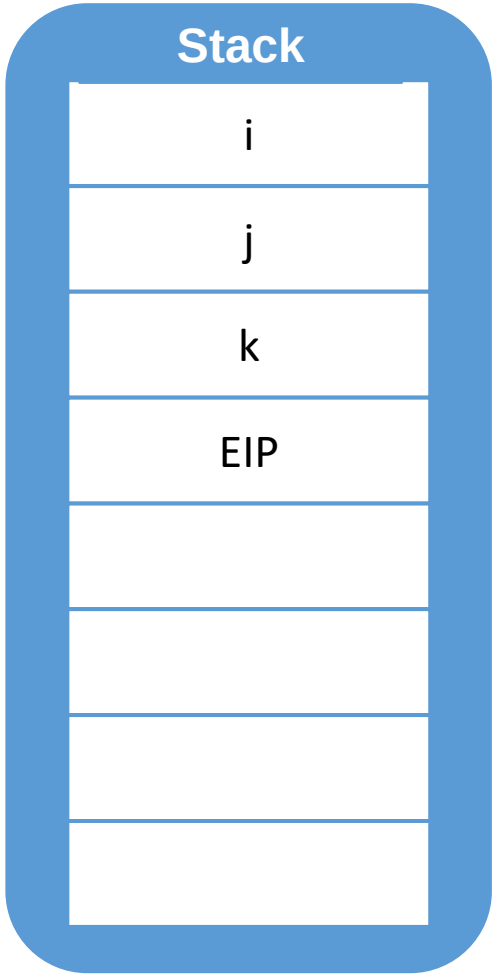
- How can we access I, J, K?
  - The first POP will extract the return address
  - RET will no longer work…

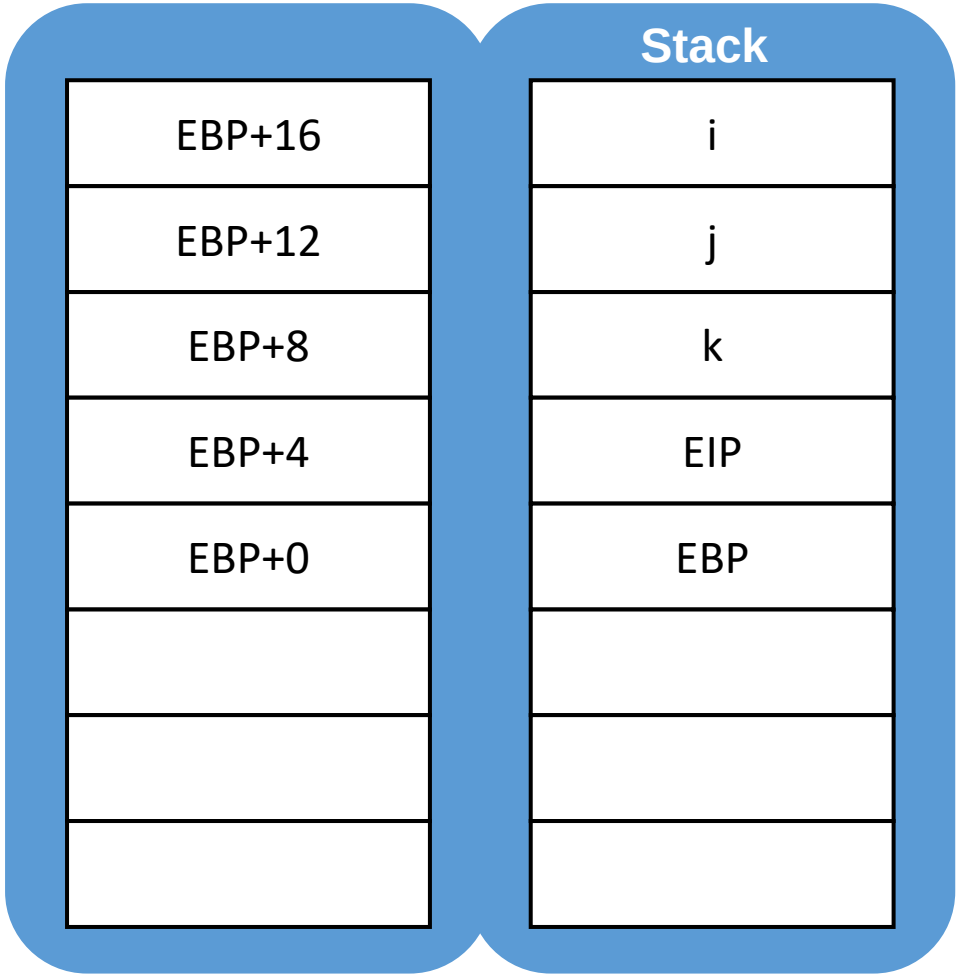


EBP

- EBP – (Extended) Base Pointer
- Assists accessing values on the stack

```
; eax = I+J-k
push    ebp
mov     ebp, esp
; ...
pop     ebp
ret     12
```

- How does it help?



EBP- cont.

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- Our code can now be written with relative addresses:

```
simple:
        ; eax = I+J-k
        push    ebp
        mov     ebp, esp
        mov     eax, [ebp+16]
        add     eax, [ebp+12]
        sub     eax, [ebp+8]
        pop     ebp
        ret     12
```

RET + Const

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- Think: Why have we used RET 12?
- RET + Const:
  - ESP = ESP+Const
  - RET
- Why 12?
  - We pushed 12 bytes (3 x 32 bit)

Local Variables

- A function may have local variables
- Stored on the stack
- Any memory reserved on the stack – reduces ESP
  - Reserve 8 bytes: `sub ESP, 8`
  - Free reserved memory: `add ESP, 8`

Local Variables – cont.

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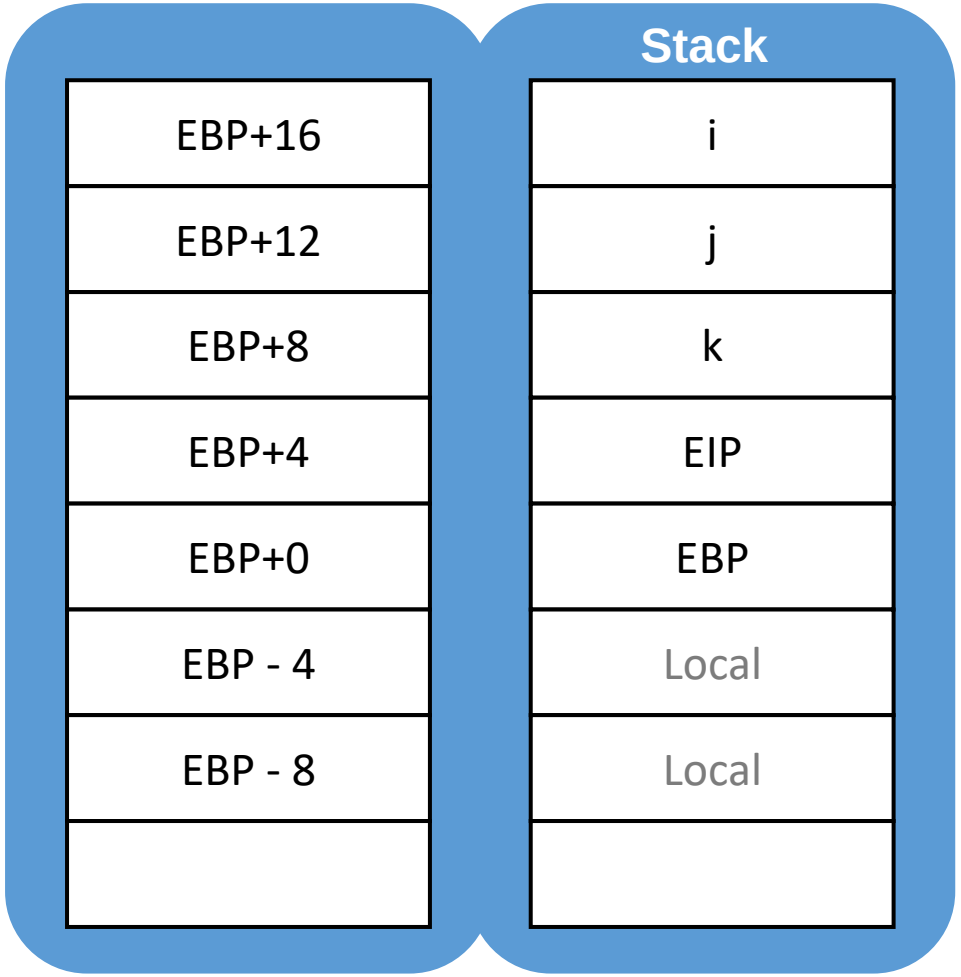
- Accessing locals- easy and fun with EBP :-)
- Don’t forget to free memory before RET
  - Or else, program might crash…



Class Exercise

- Fibon.asm

LEAVE

Equivalent to:

```
mov     esp, ebp
pop     ebp
```

What is the purpose?

Clear local vars & restore ebp with one opcode

Calling Conventions

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- Caller – the code that calls the function
- Callee – the function which is called
- Must coordinate between caller & callee:
  - Order of passing parameters to the stack
  - Register to store return value
  - “Cleaning” the stack

Calling Conventions- cont.

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- Consider the C function declaration:

  int MyFunc(int a, int b)

- And the following code:

  int result = MyFunc(1,2)

- How will the code be assembled?

```
push    1
push    2
call    MyFunc
```

```
push    2
push    1
call    MyFunc
```

Calling Conventions –cont.

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- Cleaning the stack:

```
; callee cleans stack
push    1
push    2
call    MyProc
; MyProc has RET 8
```

```
; caller cleans stack
push    1
push    2
call    MyProc
add     esp, 8
```

CDCEL / STDCALL/FASTCALL

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	CDECL	STDCALL	FASTCALL
Passing Params	Right to Left	Right to Left	Left to Right*
Return reg.	EAX	EAX	EAX
Clean the stack	Caller	Callee	Callee

* First 2 arguments, the rest are right to left

Translation to Assembly

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```
; CDECL: passing params right to left
push    2
push    1
call    MyProc
; caller cleans the stack
add     esp, 8
```

```
; STDCALL: passing params right to left
push    2
push    1
call    MyProc
; callee cleans the stack
```

```
; FASTCALL: passing params left to right, ECX & EDX
mov     ecx, 1
mov     edx, 2
call    MyProc
```

Advantages

- CDECL advantage- number of params is flexible
  - Caller knows and cleans stack
  - Example- print
- STDCALL advantage – faster
  - Callee has RET N
  - Saves one instruction (ADD ESP, N)
- FASTCALL even faster
  - Registers are faster than memory (stack)

ExitProcess

- Self study ExitProcess only through MSDN
  - What is the input parameter?
  - Which DLL file should be included?

Including DLL's

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```
section '.idata' import data readable

library kernel,'kernel32.dll'

import  kernel,\
        ExitProcess,'ExitProcess'
```

Set section

DLL to be imported

Function name

Class Exercise – call PRINTF

- Print ‘My name is %s my age is %d’

Exercise

- Add the following functionality:
  - A console message ‘Please enter text’ (printf)
  - Calculate length of user text (strlen)
  - Print the user text and it’s length (printf)

```
Please enter text
Hello Cool Cyber Class!
User entered: Hello Cool Cyber Class!
Length: 23
```

Done :-)

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```
push class
call great_work
```

QUESTIONS